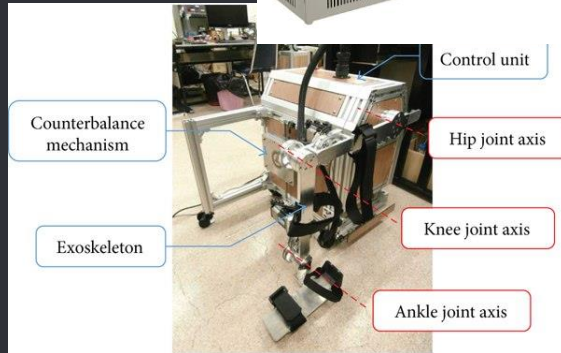




- **AI BASED KNEE
REHABILITATION MECHANISM**

INTRODUCTION



PROBLEM STATEMENT



- **India:** According to the World Health Organization (WHO), approximately 150 million people in India have some form of musculoskeletal condition, including knee abnormalities. Of these, an estimated 2.5 million people need knee rehabilitation assistance.
- **Worldwide:** The WHO estimates that 1.71 billion people worldwide have musculoskeletal conditions. Of these, an estimated 300 million people need knee rehabilitation assistance.

The prevalence of knee abnormalities is increasing worldwide due to a number of factors, including:

- **Age:** The risk of knee abnormalities increases with age.
- **Obesity:** Obesity is a major risk factor for knee problems.
- **Physical activity:** Too much or too little physical activity can both increase the risk of knee abnormalities.

OBJECTIVES



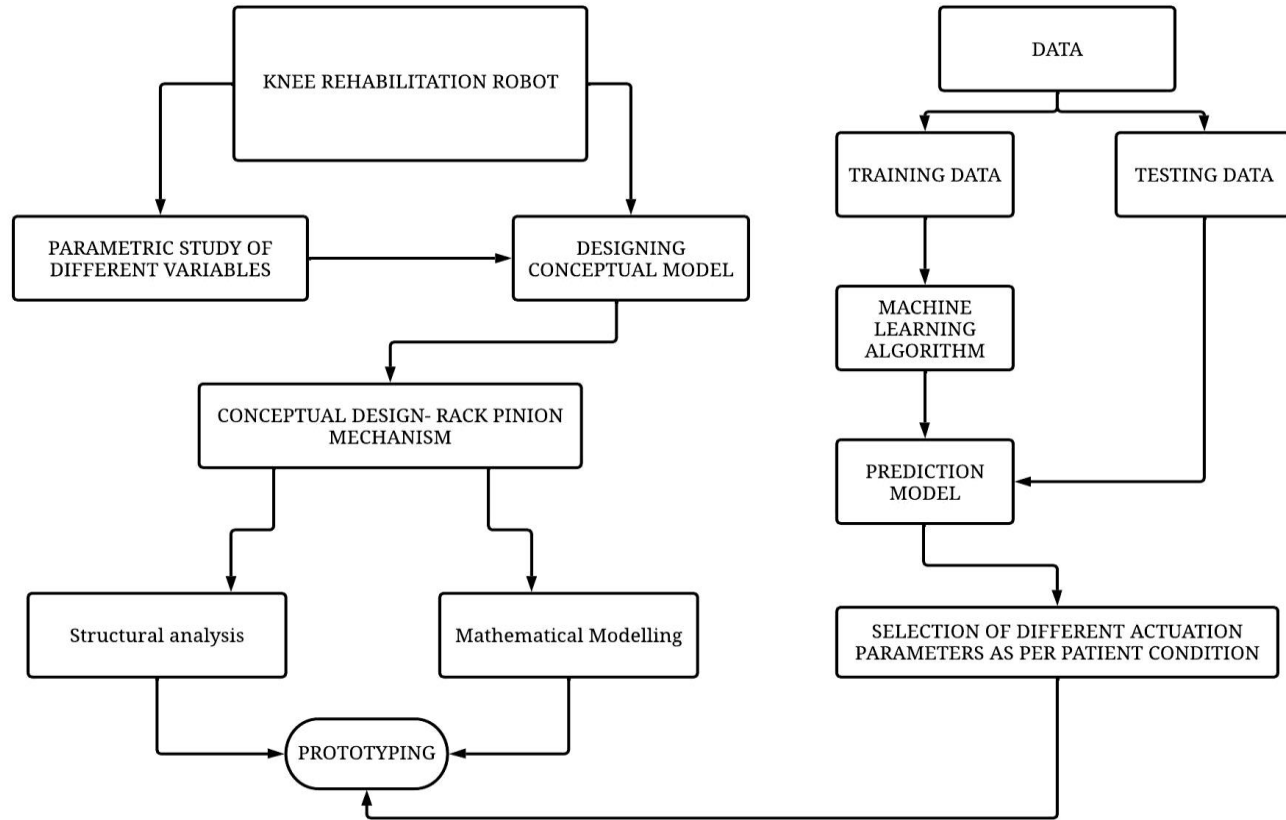
- Diagnosis of knee abnormalities using image processing and AI algorithm
- Design and simulation of the model
- Analysis of the model
- Fabrication of the mechanism

EXPECTED OUTPUT

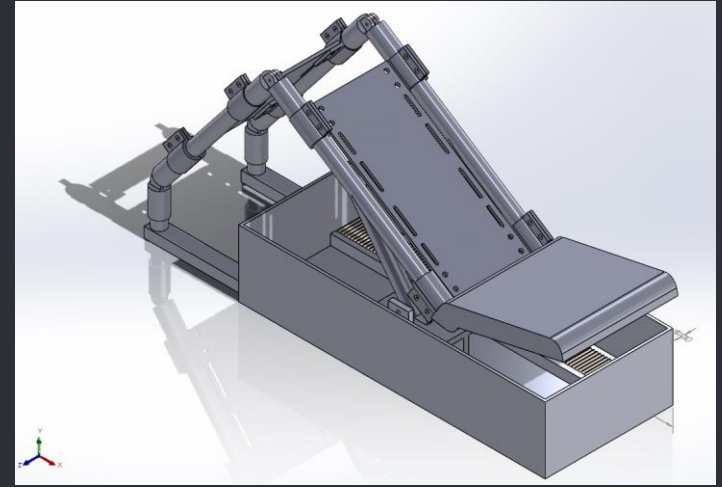
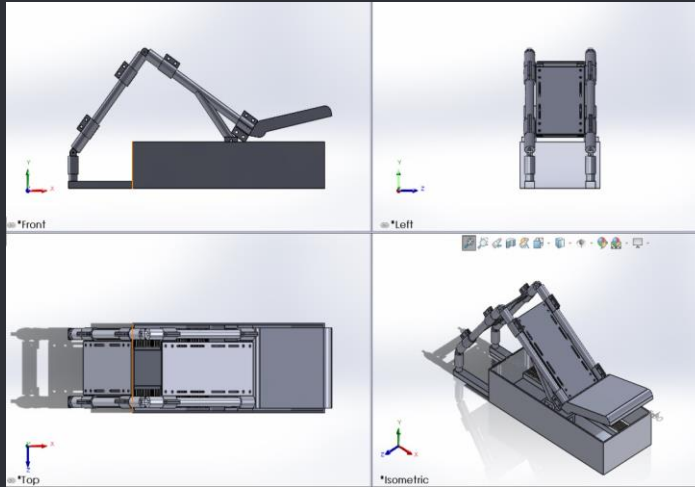


- An AI model which is able to predict/ detect any knee abnormalities found/ present.
- Design and development of mechanism for rehabilitation process.

METHODOLOGY



3D CAD MODEL



- ❑ Based on Sliding actuation using rack & pinion transmission mechanism
- ❑ Material used – Thermoplastic & Structural Steel (for joint-linkages)
- ❑ Weight of the model – 11.7 kgs

STRUCTURAL ANALYSIS

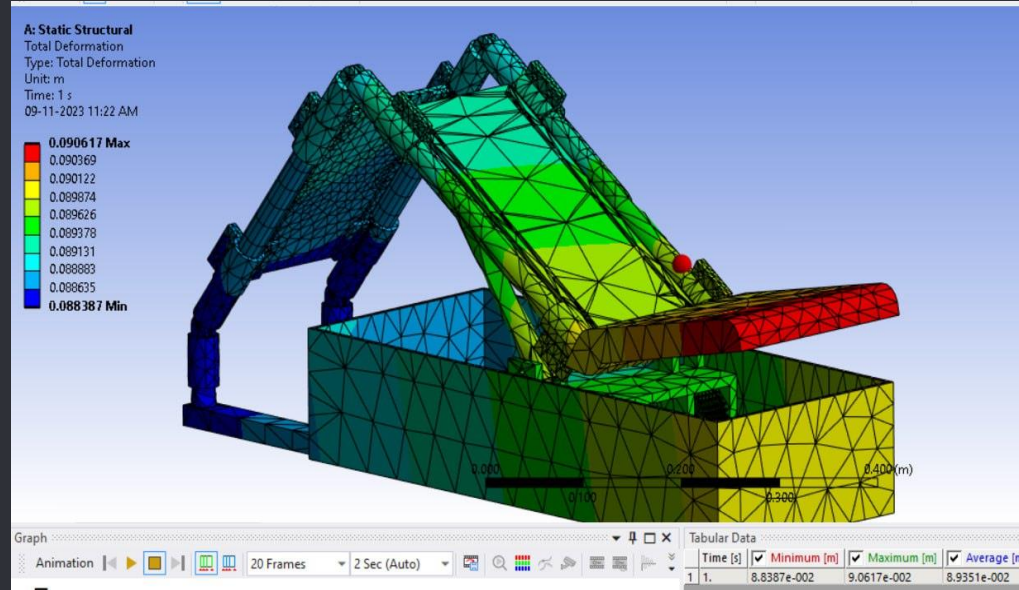


Considerations for testing:

- PI Thermoplastic, PVC, ABS material is used for the body
- structural steel is used for joints and hinges.
- The force applied on the leg supports is 150N
- Based is fixed

STRUCTURAL ANALYSIS

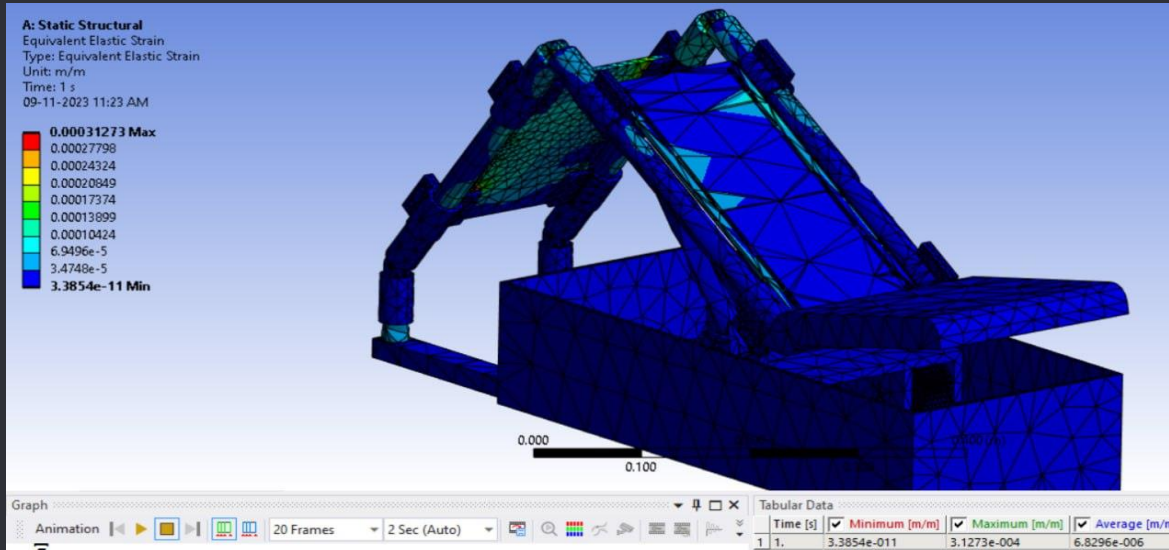
Total deformation



It was observed that deformation ranges from 0.090617m maximum on footplate and 0.088387m on the Base Joint respectively.

STRUCTURAL ANALYSIS

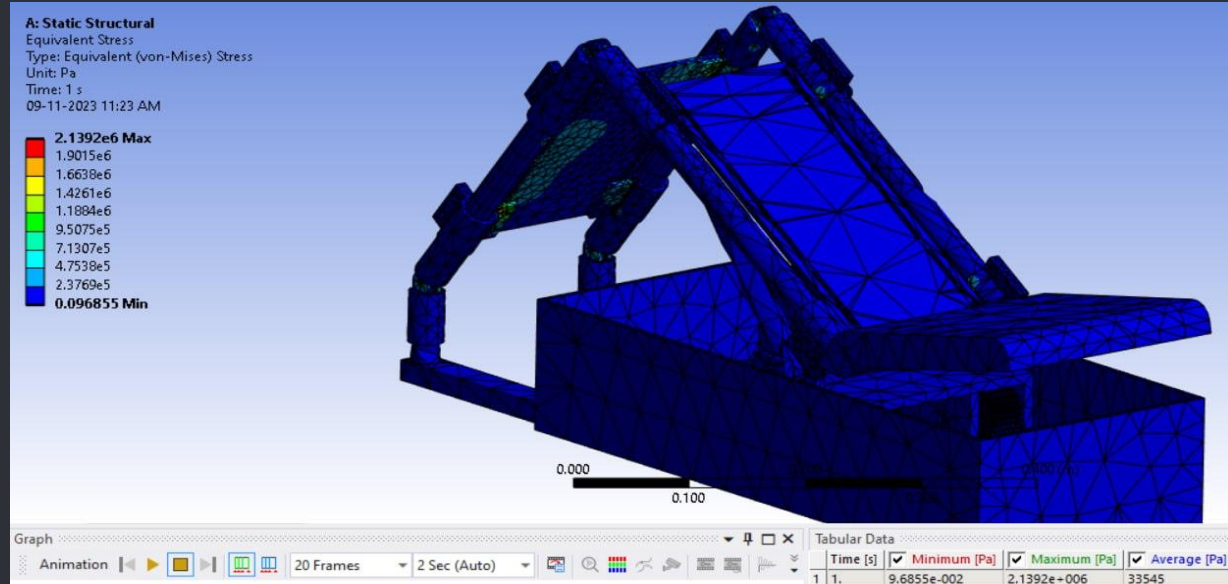
Strain



It was observed maximum strain was induced on the thigh support ranging 0.000312m and minimum on Base body of 3.3854×10^{-11} m

STRUCTURAL ANALYSIS

Stress



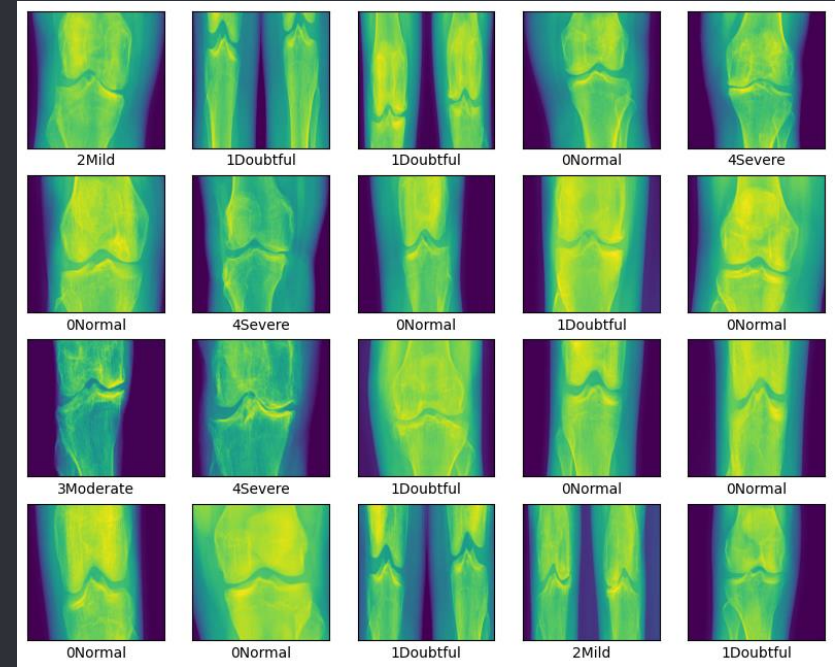
It states that maximum strain levels were observed on thigh support of $0.021392 \times 10^{-8} \text{m}$ and minimum of 0.096855m on the base body.

AI ALGORITHM OUTPUT

Different grading images to classify the knee condition for diagnosis and prediction of the need of Knee Rehabilitation



The image represents the difference in the width between the lower limb bone joints namely tibia and femur



AI ALGORITHM OUTPUT

Relu activation training model



Training Loss vs Accuracy

test loss: 3.478864908218384 %

test accuracy: 0.5575757622718811 %

True positive=92 of 162

true label	Normal	Doubtful	Mid	Moderate	Severe	
	17	2	1	5	1	
	2	29	9	5	2	
	1	12	40	4	1	
	2	6	3	4	1	
	5	9	0	2	2	
		predicted label				

Grading criteria based on confusion matrix

AI ALGORITHM OUTPUT

Linear activation function

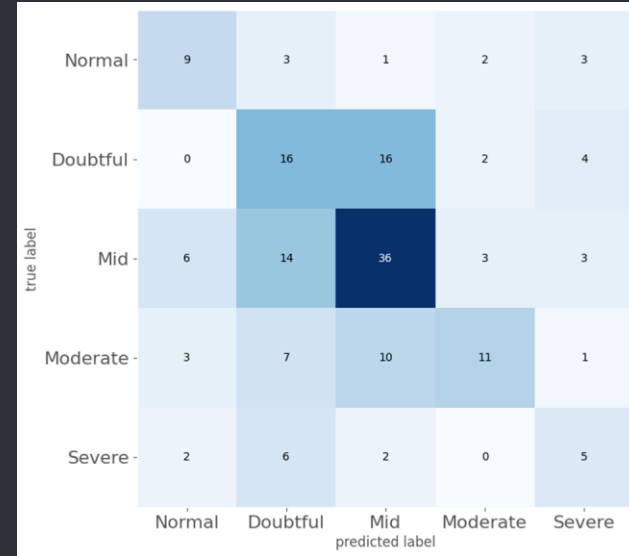


Training Loss vs Accuracy

test loss: 3.7095234394073486 %

test accuracy: 0.46666666865348816 %

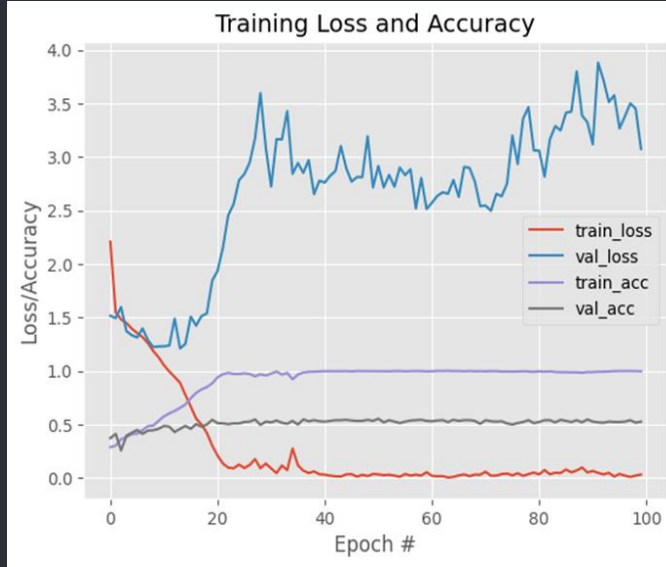
True positive=77 of 162



Grading criteria based on confusion matrix

AI ALGORITHM OUTPUT

Leaky Relu activation function



Training Loss vs Accuracy

test loss: 4.584426403045654 %

test accuracy: 0.521212100982666 %

True positive: 86 of 162

true label	Normal	Doubtful	Mid	Moderate	Severe	
	11	4	1	3	1	
	1	33	18	3	3	
	0	14	22	3	0	
	5	2	4	12	4	
	4	7	0	2	8	
		predicted label				
		Normal	Doubtful	Mid	Moderate	Severe

Grading criteria based on confusion matrix

AI ALGORITHM OUTPUT



Activation Function	Epoch	Test loss	Test accuracy
<i>Relu</i>	100	3.48	0.56
<i>Relu</i>	50	2.82	0.53
<i>Linear</i>	100	3.71	0.47
<i>Linear</i>	50	3.11	0.46
<i>LeakyRelu</i>	100	4.58	0.52
<i>LeakyRelu</i>	50	3.08	0.50

Source Code Link -

https://colab.research.google.com/drive/1uN9005P1ps0T7iTNDYV62Jl_ArKktOZW#scrollTo=XoKGbeNYJXF

● CONCLUSION

Prediction part

- The model is created by comparing different activation functions.
- The difference in accuracy ranges from 2-5% by changing the activation made to choose Relu over linear and leakyrelu.
- The final accuracy observed was 55% with the deep-learning model. The images chosen—around 169—were validated alone.

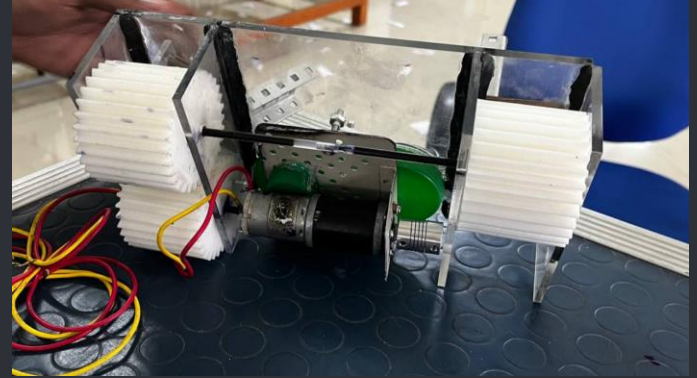
Design part

- The model is created by comparing different linear actuation mechanisms.
- A combination of RPRPR joint configuration is used which is actuated with the help of pinions running on rack connected at the end of the configuration.

● PROTOTYPE



LINKAGES ASSEMBLY



PINNION ASSEMBLY



EXTENDED



CONTRACTED

LITERATURE REVIEW



Title	Author	Inference	Methodology
Development And Control Of A New Sitting-type Lower Limb Rehabilitation Robot, 2017	J.K. Mohan , S. Mohan , P. Deepasundar , R. Kirubashankar	Use of prismatic joints for smooth limb motion Use of parallel manipulators for lower limb actuation	Kinematic and dynamic analyses PID control scheme
Knee rehabilitation using an intelligent robotic system, 2009	Akdoğan, E., Taçgın, E., & Adli, M. A.	The robot manipulator works based on impedance control which is a control method for physiotherapy	The System has a servo motor and its driver as actuator, force/torque sensor and its controller for measurement of force data that come from therapist and patient and a data acquisition card

LITERATURE REVIEW

Title

Author

Inference

Methodology

Knee rehabilitation device with soft actuation: An approach to the motion control, 2018

Solaque, L., Romero, M., & Velasco

The feedforward strategy compensates loading disturbances while the feedback strategy acts in a defined threshold to maintain the stiffness of the system.

This paper tackles an approach to controlling a soft actuated rehabilitation device.

It aims to control the link angular position of the knee rehabilitation device, to comply with defined repetitive motions

Fundamental Concepts of Convolutional Neural Network.

Ghosh, Anirudha & Sufian, A. & Sultana, Farhana & Chakrabarti, Amlan & De, Debashis.

This CNN has been using to build many use cases models from digit recognition to complex medical image analysis.

This paper tried to explain each components of a CNN, how it works to image analysis, and other relevant things.

This paper gives a review from foundation of CNN to latest models and mentioned some applications areas

The research methodology involves a thorough review of Convolutional Neural Networks (CNNs), tracing their evolution, technical components, and key models. It explores fundamental aspects like convolutional kernels, operations, and activation functions, offering insights into CNNs' applications in computer vision and natural language processing.

LITERATURE REVIEW

Title	Author	Inference	Methodology
Design and control of a lower limb rehabilitation robot considering undesirable torques of the patient's limb.	Almaghout, K., Tarvirdizadeh, B., Alipour, K., & Hadi, A.	The robot has a force controller unit with two controllers one for each leg. Using two controllers helps to receive data from two force sensors to operate quickly when controlling four motors	The robot uses the linear motion mechanisms in which the robot lifts and lowers the legs by kinematic interpretation, and the force sensor is attached to a device for pushing up the calf and thigh, measuring the pushing force, including the weight of the lower limb.
Rehabilitation of total knee arthroplasty by integrating conjoint isometric myodynamia and real-time rotation sensing system.	Luo, J., Li, Y., He, M., Wang, Z., Li, C., Liu, D., An, J., Xie, W., He, Y., Xiao, W., Li, Z., Wang, Z. L., & Tang, W.	Two sensing modules are mounted onto braces, 1) isometric myodynamia measurement (real-time force analysis) 2) active range of joint movement measurement (real-time angle analysis).	Use Active Rotation Sensing Module, Isometric Myodynamia Measurement Module and effectively get data online

LITERATURE REVIEW

Title

The use of robotics devices in knee rehabilitation: a critical review.

Author

Wilmart, R., Garone, E., & Innocenti, B.

Inference

Advantages and different types of the robots used for rehabilitation.

Methodology

By using Continuous Passive Motion (CPM) machines and Therapeutic Exercise Machines (TEM);

- Gait-training Robots;
- Exoskeletons.

Deep learning-based algorithm for assessment of knee osteoarthritis severity in radiographs matches performance of radiologists

Swiecicki, A., Li, N., O'Donnell, J., Said, N., Yang, J., Mather, R. C., Jiranek, W. A., & Mazurowski, M. A.

The image prediction is given while using only classification which achieved 61% whereas the highest of 71.9 achieved by using PA and LAT channels which other combinations are also given

The prediction of knee osteoarthritis is done in different models from normal classification, regression to PA and LAT channels

LITERATURE REVIEW



Title	Author	Inference	Methodology
Artificial intelligence in diagnosis of knee osteoarthritis and prediction of arthroplasty outcomes: a review.	Lee, L.S., Chan, P.K., Wen, C. et al.	Artificial intelligence offers a promising solution to improve detection and management of knee osteoarthritis. Further research to overcome the weaknesses of machine learning models may enhance reliability and allow for future use in routine healthcare settings.	Using specific terms related to AI, ML, knee osteoarthritis, and arthroplasty. Inclusion criteria targeted studies on AI in KOA diagnosis, predicting TKA need, and TKA outcomes. A total of 136 results were screened, with 22 papers meeting the inclusion criteria. The evaluation focused on strengths, weaknesses, and the need for external validation of AI algorithms
Machine learning in knee osteoarthritis: A review	Kokkotis, C., Moustakidis, S., Papageorgiou, E., Giakas, G., & Tsaopoulos, D. E.	The well-defined report of Xray vs MRI along with regression and other ML techniques are defined with accuracy to understand the features.	The different machine learning algorithms are compared and the advantages and disadvantages are given clearly

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THANKS!